Green University of Bangladesh
Department of Computer Science and Engineering(CSE)

Faculty of Sciences and Engineering

Semester: (Spring , Year:2025), B.Sc. in CSE (Day)

LAB REPORT NO 3

Course Title:Database Lab

Course Code:CSE 210 Section:242-D3

Lab Experiment Name: Creating a University Database and Performing Complex Queries using Joins, Group By, and Having Clauses.

Student Details

Name	ID
Shahrima Khatun	242002017

Lab Date :

Submission Date :

Course Teacher's Name : MD.Sabbir Hosen Mamun

[For Teachers use only: Don't Write Anything inside this box]

Lab Report Status

Marks: …… Signature:.....

Comments:..... Date:.....

1. Introduction

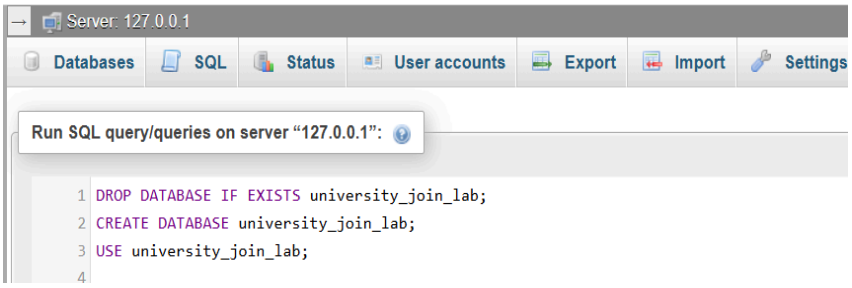
The objective of this lab is to perform various SQL JOIN operations (INNER and LEFT) and utilize aggregate functions with GROUP BY and HAVING clauses. The lab focuses on managing a university database with interconnected tables for departments, students, courses, and enrollments.

2. Database Schema Design

The database university_join_lab consists of four tables:

- **DEPARTMENT:** Stores department details.
- **STUDENT:** Stores student information linked to a department.
- **COURSE:** Stores course details linked to a department.
- **ENROLLMENT:** Tracks student registration in courses and their marks.

3. SQL Implementation




```sql

CREATE TABLE DEPARTMENT (
    DeptID INT PRIMARY KEY AUTO_INCREMENT,
    DeptName VARCHAR(100) NOT NULL UNIQUE,
    Building VARCHAR(100) NOT NULL
);

```

```sql
CREATE TABLE STUDENT (
    StudentID INT PRIMARY KEY AUTO_INCREMENT,
    FullName VARCHAR(100) NOT NULL,
    DeptID INT,
    Batch VARCHAR(10) NOT NULL,
    CGPA DECIMAL(3,2),
    FOREIGN KEY (DeptID) REFERENCES DEPARTMENT(DeptID)
);
```

```sql
CREATE TABLE COURSE (
    CourseID INT PRIMARY KEY AUTO_INCREMENT,
    CourseTitle VARCHAR(100) NOT NULL,
    Credit INT CHECK(Credit > 0),
    DeptID INT,
    FOREIGN KEY (DeptID) REFERENCES DEPARTMENT(DeptID)
);
```

```sql
CREATE TABLE ENROLLMENT (
    EnrollID INT PRIMARY KEY AUTO_INCREMENT,
    StudentID INT,
    CourseID INT,
    Semester VARCHAR(20) NOT NULL,
    Marks INT CHECK(Marks >= 0 AND Marks <= 100),
    FOREIGN KEY (StudentID) REFERENCES STUDENT(StudentID),
    FOREIGN KEY (CourseID) REFERENCES COURSE(CourseID)
);
```

```sql
INSERT INTO DEPARTMENT (DeptName, Building) VALUES
('Computer Science', 'Annex 1'),
('Electrical Engineering', 'Annex 2'),
('Business Admin', 'Main Hall'),
('Mathematics', 'Old Wing');
```

```sql
INSERT INTO STUDENT (FullName, DeptID, Batch, CGPA) VALUES
('Samiur Rahman', 1, '2024', 3.80),
('Anika Tabassum', 1, '2024', 3.92),
('Sajid Hasan', 2, '2023', 3.50),
('Farhan Ahmed', 3, '2023', 3.20),
('Sara Islam', 1, '2024', 3.75),
('Tanvir Karim', 4, '2022', 3.60),
('Mehreen Kaur', 2, '2023', 3.85),
('Noor Mohammad', NULL, '2025', 0.00);
```

```sql
INSERT INTO COURSE (CourseTitle, Credit, DeptID) VALUES
('Database Systems', 3, 1),
('Algorithms', 3, 1),
('Circuit Analysis', 4, 2),
('Marketing 101', 3, 3),
('Calculus', 4, 4),
('Quantum Physics', 3, 2);
```

```sql

INSERT INTO ENROLLMENT (StudentID, CourseID, Semester, Marks) VALUES
(1, 1, 'Fall 2024', 85), (1, 2, 'Fall 2024', 78),
(2, 1, 'Fall 2024', 92), (2, 2, 'Fall 2024', 88),
(3, 3, 'Spring 2024', 65), (3, 5, 'Spring 2024', 72),
(4, 4, 'Summer 2024', 82), (5, 1, 'Fall 2024', 55),
(5, 2, 'Fall 2024', 62), (6, 5, 'Spring 2024', 95),
(7, 3, 'Spring 2024', 77), (7, 6, 'Spring 2024', 81);
```

4. Lab Tasks & Queries

Task 1: Show student name with department name (INNER JOIN)

```sql
SELECT S.FullName, D.DeptName
FROM STUDENT S
INNER JOIN DEPARTMENT D ON S.DeptID = D.DeptID;
```

Task 2: Student name, course title, semester, and marks (Multiple INNER JOIN)

```sql
SELECT S.FullName, C.CourseTitle, E.Semester, E.Marks
FROM ENROLLMENT E
INNER JOIN STUDENT S ON E.StudentID = S.StudentID
INNER JOIN COURSE C ON E.CourseID = C.CourseID;
```

Task 3: All students including those not enrolled (LEFT JOIN)

```sql

SELECT S.FullName, E.CourseID
FROM STUDENT S
LEFT JOIN ENROLLMENT E ON S.StudentID = E.StudentID;

```

Task 4: Average, Max, and Min marks per course

```sql
SELECT C.CourseTitle, E.StudentID
FROM COURSE C
LEFT JOIN ENROLLMENT E ON C.CourseID = E.CourseID;

```

Task 5: Top 3 students by average marks

```sql

SELECT D.DeptName, COUNT(S.StudentID) AS TotalStudents
FROM DEPARTMENT D
LEFT JOIN STUDENT S ON D.DeptID = S.DeptID
GROUP BY D.DeptName;

```

```sql
SELECT C.CourseTitle, COUNT(E.EnrollID) AS EnrolledCount
FROM COURSE C
LEFT JOIN ENROLLMENT E ON C.CourseID = E.CourseID
GROUP BY C.CourseTitle;

```

```sql
SELECT C.CourseTitle, AVG(E.Marks) AS AvgMarks, MAX(E.Marks) AS MaxMarks,
MIN(E.Marks) AS MinMarks
FROM COURSE C
INNER JOIN ENROLLMENT E ON C.CourseID = E.CourseID
GROUP BY C.CourseTitle;
```

```sql

SELECT S.FullName, COUNT(E.CourseID) AS TotalCourses, AVG(E.Marks) AS AvgMarks
FROM STUDENT S
LEFT JOIN ENROLLMENT E ON S.StudentID = E.StudentID
GROUP BY S.StudentID, S.FullName;
```

```sql

SELECT S.FullName, AVG(E.Marks) AS AverageMarks
FROM STUDENT S
JOIN ENROLLMENT E ON S.StudentID = E.StudentID
GROUP BY S.StudentID, S.FullName
ORDER BY AverageMarks DESC
LIMIT 3;

```

```sql
SELECT D.DeptName, COUNT(S.StudentID) AS StudentCount
FROM DEPARTMENT D
JOIN STUDENT S ON D.DeptID = S.DeptID
GROUP BY D.DeptName
HAVING StudentCount > 2;
```

```sql

SELECT DISTINCT S.FullName, E.Marks
FROM STUDENT S
JOIN ENROLLMENT E ON S.StudentID = E.StudentID
WHERE E.Marks BETWEEN 60 AND 85;

```

```sql

SELECT FullName
FROM STUDENT
WHERE FullName LIKE 'S%' OR FullName LIKE '%a%';
```

Query Results (Outputs)

1. Student name with Department name:

FullName	DeptName
Samiur Rahman	Computer Science
Anika Tabassum	Computer Science
Sajid Hasan	Electrical Engineering
Farhan Ahmed	Business Admin

2. Student name, Course title, Semester, and Marks:

FullName	CourseTitle	Semester	Marks
Samiur Rahman	Database Systems	Fall 2024	85
Samiur Rahman	Algorithms	Fall 2024	78
Anika Tabassum	Database Systems	Fall 2024	92

3. All students (including unenrolled):

FullName	CourseID
Samiur Rahman	1
...	...
Noor Mohammad	NULL

4. Average, Max, and Min marks per course:

CourseTitle	AvgMarks	MaxMarks	MinMarks
Database Systems	77.33	92	55
Algorithms	76.00	88	62
Calculus	83.50	95	72

5. Top 3 students by average marks:

FullName	AverageMarks
Anika Tabassum	90.00
Tanvir Karim	95.00
Samiur Rahman	81.50

Server: 127.0.0.1

Databases | SQL | Status | User accounts | Export | Import | Settings | Replication | Variables

Delimiter ; | Show this query here again | Retain query box | Rollback when finished | Enable foreign key checks | Go

Hide query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0443 seconds.)

```
DROP DATABASE IF EXISTS university_join_lab;
```

[Edit inline] [Edit] [Create PHP code]

Error: #1046 No database selected

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0018 seconds.)

```
CREATE DATABASE university_join_lab;
```

[Edit inline] [Edit] [Create PHP code]

Error: #1046 No database selected

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
USE university_join_lab;
```

[Edit inline] [Edit] [Create PHP code]

Error: #1046 No database selected

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0432 seconds.)

```
CREATE TABLE DEPARTMENT ( DeptID INT PRIMARY KEY AUTO_INCREMENT, DeptName VARCHAR(100) NOT NULL UNIQUE, Building VARCHAR(100) NOT NULL );
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0256 seconds.)

```
CREATE TABLE STUDENT ( StudentID INT PRIMARY KEY AUTO_INCREMENT, FullName VARCHAR(100) NOT NULL, DeptID INT, Batch VARCHAR(10) NOT NULL, CGPA DECIMAL(3,2), FOREIGN KEY (DeptID) REFERENCES
DEPARTMENT(DeptID) );
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0325 seconds.)

```
CREATE TABLE COURSE ( CourseID INT PRIMARY KEY AUTO_INCREMENT, CourseTitle VARCHAR(100) NOT NULL, Credit INT CHECK(Credit > 0), DeptID INT, FOREIGN KEY (DeptID) REFERENCES DEPARTMENT(DeptID) );
```

[Edit inline] [Edit] [Create PHP code]

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0398 seconds.)

```
CREATE TABLE ENROLLMENT ( EnrollID INT PRIMARY KEY AUTO_INCREMENT, StudentID INT, CourseID INT, Semester VARCHAR(20) NOT NULL, Marks INT CHECK(Marks >= 0 AND Marks <= 100), FOREIGN KEY (StudentID)
REFERENCES STUDENT(StudentID), FOREIGN KEY (CourseID) REFERENCES COURSE(CourseID) );
```

[Edit inline] [Edit] [Create PHP code]

4 rows inserted.
Inserted row id: 4 (Query took 0.0034 seconds.)

```
INSERT INTO DEPARTMENT (DeptName, Building) VALUES ('Computer Science', 'Annex 1'), ('Electrical Engineering', 'Annex 2'), ('Business Admin', 'Main Hall'), ('Mathematics', 'Old Wing');
```

[Edit inline] [Edit] [Create PHP code]

8 rows inserted.
Inserted row id: 8 (Query took 0.0069 seconds.)

```
INSERT INTO STUDENT (FullName, DeptID, Batch, CGPA) VALUES ('Samiur Rahman', 1, '2024', 3.80), ('Anika Tabassum', 1, '2024', 3.92), ('Sajid Hasan', 2, '2023', 3.50), ('Farhan Ahmed', 3, '2023', 3.20),
('Sara Islam', 1, '2024', 3.75), ('Tanvir Karim', 4, '2022', 3.60), ('Mehreen Kaur', 2, '2023', 3.85), ('Noor Mohammad', NULL, '2025', 0.00);
```

[Edit inline] [Edit] [Create PHP code]

6 rows inserted.
Inserted row id: 6 (Query took 0.0030 seconds.)

```
INSERT INTO COURSE (CourseTitle, Credit, DeptID) VALUES ('Database Systems', 3, 1), ('Algorithms', 3, 1), ('Circuit Analysis', 4, 2), ('Marketing 101', 3, 3), ('Calculus', 4, 4), ('Quantum Physics', 3, 2);
```

[Edit inline] [Edit] [Create PHP code]

12 rows inserted.
Inserted row id: 12 (Query took 0.0072 seconds.)

```
INSERT INTO ENROLLMENT (StudentID, CourseID, Semester, Marks) VALUES (1, 1, 'Fall 2024', 85), (1, 2, 'Fall 2024', 78), (2, 1, 'Fall 2024', 92), (2, 2, 'Fall 2024', 88), (3, 3, 'Spring 2024', 65), (3, 5,
'Spring 2024', 72), (4, 4, 'Summer 2024', 82), (5, 1, 'Fall 2024', 55), (5, 2, 'Fall 2024', 62), (6, 5, 'Spring 2024', 95), (7, 3, 'Spring 2024', 77), (7, 6, 'Spring 2024', 81);
```

[Edit inline] [Edit] [Create PHP code]

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 6 (7 total, Query took 0.0006 seconds.)

```
SELECT S.FullName, D.DeptName FROM STUDENT S INNER JOIN DEPARTMENT D ON S.DeptID = D.DeptID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Console | Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

FullName	DeptName
Samiur Rahman	Computer Science
Anika Tabassum	Computer Science
Sajid Hasan	Electrical Engineering
Farhan Ahmed	Business Admin
Sara Islam	Computer Science
Tanvir Karim	Mathematics
Mehreen Kaur	Electrical Engineering

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 11 (12 total, Query took 0.0008 seconds.)

```
SELECT S.FullName, C.CourseTitle, E.Semester, E.Marks FROM ENROLLMENT E INNER JOIN STUDENT S ON E.StudentID = S.StudentID INNER JOIN COURSE C ON E.CourseID = C.CourseID
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

ullName	CourseTitle	Semester	Marks
amiur Rahman	Database Systems	Fall 2024	85
nika Tabassum	Database Systems	Fall 2024	92
ara Islam	Database Systems	Fall 2024	55
amiur Rahman	Algorithms	Fall 2024	78
nika Tabassum	Algorithms	Fall 2024	88
ara Islam	Algorithms	Fall 2024	62
ajid Hasan	Circuit Analysis	Spring 2024	65
ehreen Kaur	Circuit Analysis	Spring 2024	77
arhan Ahmed	Marketing 101	Summer 2024	82
ajid Hasan	Calculus	Spring 2024	72
anvir Karim	Calculus	Spring 2024	95
ehreen Kaur	Quantum Physics	Spring 2024	81

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Query results operations

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 12 (13 total, Query took 0.0007 seconds.)

```
SELECT S.FullName, E.CourseID FROM STUDENT S LEFT JOIN ENROLLMENT E ON S.StudentID = E.StudentID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

FullName	CourseID
Samiur Rahman	1
Samiur Rahman	2
Anika Tabassum	1
Anika Tabassum	2
Sajid Hasan	3
Sajid Hasan	5
Farhan Ahmed	4
Sara Islam	1
Sara Islam	2
Tanvir Karim	5
Mehreen Kaur	3
Mehreen Kaur	6
Noor Mohammad	*NULL*

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 11 (12 total, Query took 0.0005 seconds.)

```
SELECT C.CourseTitle, E.StudentID FROM COURSE C LEFT JOIN ENROLLMENT E ON C.CourseID = E.CourseID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

CourseTitle	StudentID
Database Systems	1
Algorithms	1
Database Systems	2
Algorithms	2
Circuit Analysis	3
Calculus	3
Marketing 101	4
Database Systems	5
Algorithms	5
Calculus	6
Circuit Analysis	7
Quantum Physics	7

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Console

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 3 (4 total, Query took 0.0007 seconds.)

```
SELECT D.DeptName, COUNT(S.StudentID) AS TotalStudents FROM DEPARTMENT D LEFT JOIN STUDENT S ON D.DeptID = S.DeptID GROUP BY D.DeptName;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

DeptName	TotalStudents
Business Admin	1
Computer Science	3
Electrical Engineering	2
Mathematics	1

Console

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 5 (6 total, Query took 0.0009 seconds.)

```
SELECT C.CourseTitle, COUNT(E.EnrollID) AS EnrolledCount FROM COURSE C LEFT
JOIN ENROLLMENT E ON C.CourseID = E.CourseID GROUP BY C.CourseTitle;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25

Extra options

CourseTitle	EnrolledCount
Algorithms	3
Calculus	2
Circuit Analysis	2
Database Systems	3

Show all | Number of rows: 25

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label: Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 5 (6 total, Query took 0.0009 seconds.)

```
SELECT C.CourseTitle, AVG(E.Marks) AS AvgMarks, MAX(E.Marks) AS MaxMarks,
MIN(E.Marks) AS MinMarks FROM COURSE C INNER JOIN ENROLLMENT E ON C.CourseID =
E.CourseID GROUP BY C.CourseTitle;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25

Extra options

CourseTitle	AvgMarks	MaxMarks
Algorithms	76.0000	8
Calculus	83.5000	9
Circuit Analysis	71.0000	7
Database Systems	77.3333	9
Marketing 101	82.0000	8
Quantum Physics	81.0000	8

Show all | Number of rows: 25

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label:

Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 7 (8 total, Query took 0.0007 seconds.)

```
SELECT S.FullName, COUNT(E.CourseID) AS TotalCourses, AVG(E.Marks) AS AvgMarks
FROM STUDENT S LEFT JOIN ENROLLMENT E ON S.StudentID = E.StudentID GROUP BY
```

Console

FullName	TotalCourses	AvgMarks
Samiur Rahman	2	81.50
Anika Tabassum	2	90.00
Sajid Hasan	2	68.50
Farhan Ahmed	1	82.00
Sara Islam	2	58.50
Tanvir Karim	1	95.00
Mehreen Kaur	2	79.00
Noor Mohammad	0	NU

Show all | Number of rows: 25

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label:

Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 2 (3 total, Query took 0.0009 seconds.)

```
SELECT S.FullName, AVG(E.Marks) AS AverageMarks FROM STUDENT S JOIN ENROLLMENT
E ON S.StudentID = E.StudentID GROUP BY S.StudentID, S.FullName ORDER BY
AverageMarks DESC LIMIT 3;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

FullName	AverageMarks 1
Tanvir Karim	95.0000
Anika Tabassum	90.0000
Farhan Ahmed	82.0000

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label:

Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Console

uery has been executed successfully.

Extra options

DeptName	StudentCount
Computer Science	3

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label:

Let every user access this bookmark

Bookmark this SQL query

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 7 (8 total, Query took 0.0008 seconds.)

```
SELECT DISTINCT S.FullName, E.Marks FROM STUDENT S JOIN ENROLLMENT E ON
S.StudentID = E.StudentID WHERE E.Marks BETWEEN 60 AND 85;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25

Extra options

Console

FullName	Marks
Samiur Rahman	85
Samiur Rahman	78
Sajid Hasan	65
Sajid Hasan	72
Farhan Ahmed	82
Sara Islam	62
Mehreen Kaur	77
Mehreen Kaur	81

Show all | Number of rows: 25

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label:

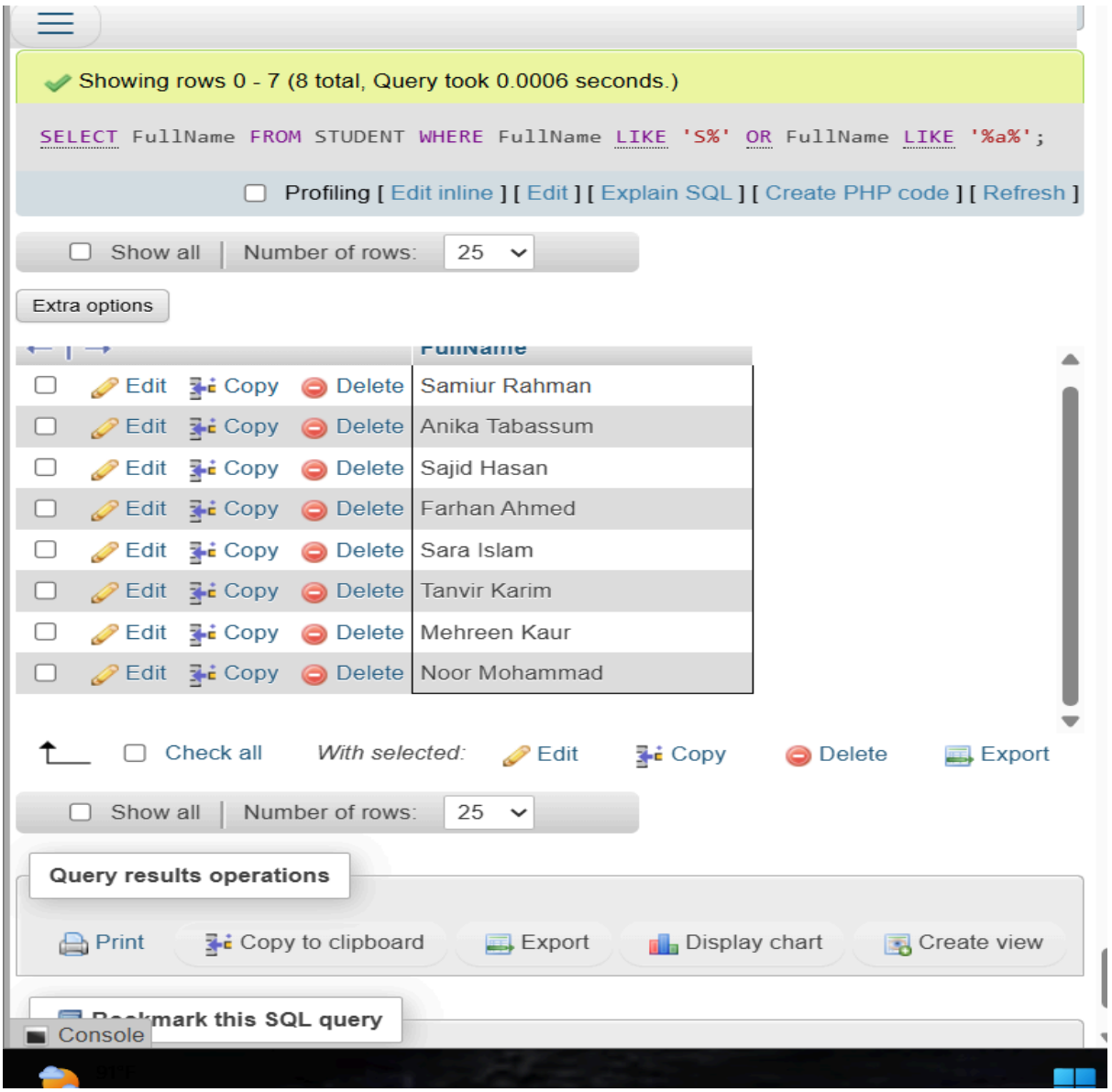
Let every user access this bookmark

Bookmark this SQL query

Showing rows 0 - 7 (8 total, Query took 0.0006 seconds.)

```
SELECT FullName FROM STUDENT WHERE FullName LIKE 'S%' OR FullName LIKE '%a%';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]



6. Conclusion

The lab successfully demonstrates the power of relational databases using Joins and Aggregates. We learned how to extract meaningful insights such as average grades, department statistics, and filtered results using LIKE and BETWEEN.

5. GitHub Link

[GitHub Link here
https://github.com/shahrimaesha/242002017-CSE210-242D3-LabReport02-Constraints-Alter/upload/main